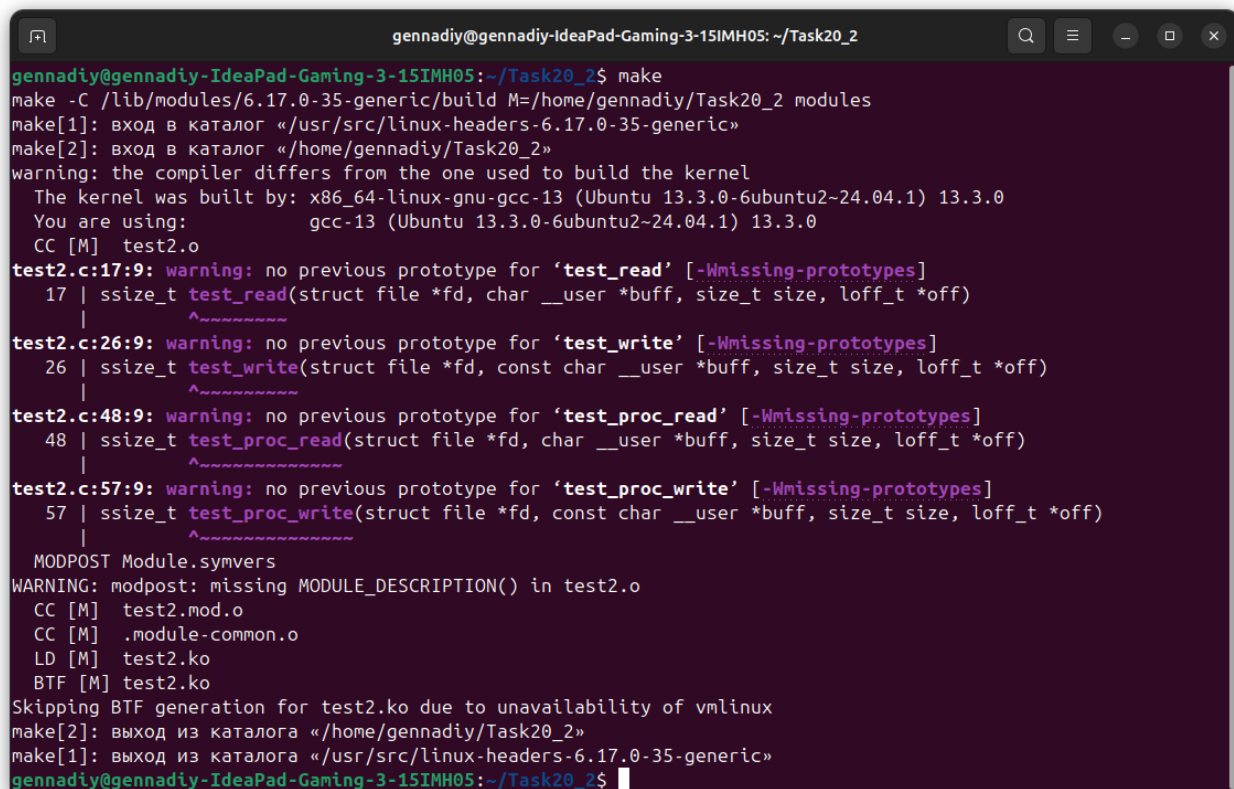
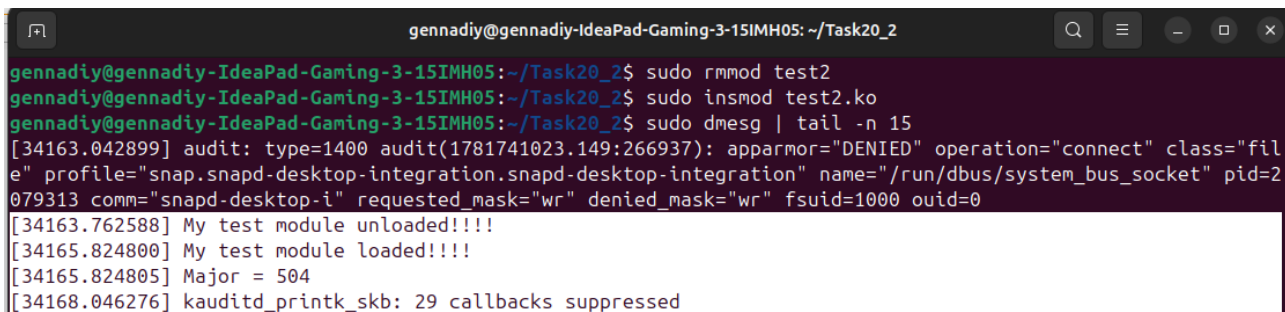


Задание 20 — модули ядра (часть 2, procfs и sysfs)



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/Task20_2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/gennadiy/Task20_2 modules
make[1]: вход в каталог «/usr/src/linux-headers-6.17.0-35-generic»
make[2]: вход в каталог «/home/gennadiy/Task20_2»
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using:
gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M] test2.o
test2.c:17:9: warning: no previous prototype for 'test_read' [-Wmissing-prototypes]
17 | ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
    |
test2.c:26:9: warning: no previous prototype for 'test_write' [-Wmissing-prototypes]
26 | ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
    |
test2.c:48:9: warning: no previous prototype for 'test_proc_read' [-Wmissing-prototypes]
48 | ssize_t test_proc_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
    |
test2.c:57:9: warning: no previous prototype for 'test_proc_write' [-Wmissing-prototypes]
57 | ssize_t test_proc_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
    |
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test2.o
CC [M] test2.mod.o
CC [M] .module-common.o
LD [M] test2.ko
BTF [M] test2.ko
Skipping BTF generation for test2.ko due to unavailability of vmlinux
make[2]: выход из каталога «/home/gennadiy/Task20_2»
make[1]: выход из каталога «/usr/src/linux-headers-6.17.0-35-generic»
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$
```

Рисунок 1 — результаты сборки



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/Task20_2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo rmmod test2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo insmod test2.ko
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo dmesg | tail -n 15
[34163.042899] audit: type=1400 audit(1781741023.149:266937): apparmor="DENIED" operation="connect" class="file" profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/run/dbus/system_bus_socket" pid=2079313 comm="snapd-desktop-i" requested_mask="wr" denied_mask="wr" fsuid=1000 ouid=0
[34163.762588] My test module unloaded!!!!
[34165.824800] My test module loaded!!!!
[34165.824805] Major = 504
[34168.046276] kauditd_printk_skb: 29 callbacks suppressed
```

Рисунок 2 — лог при загрузке

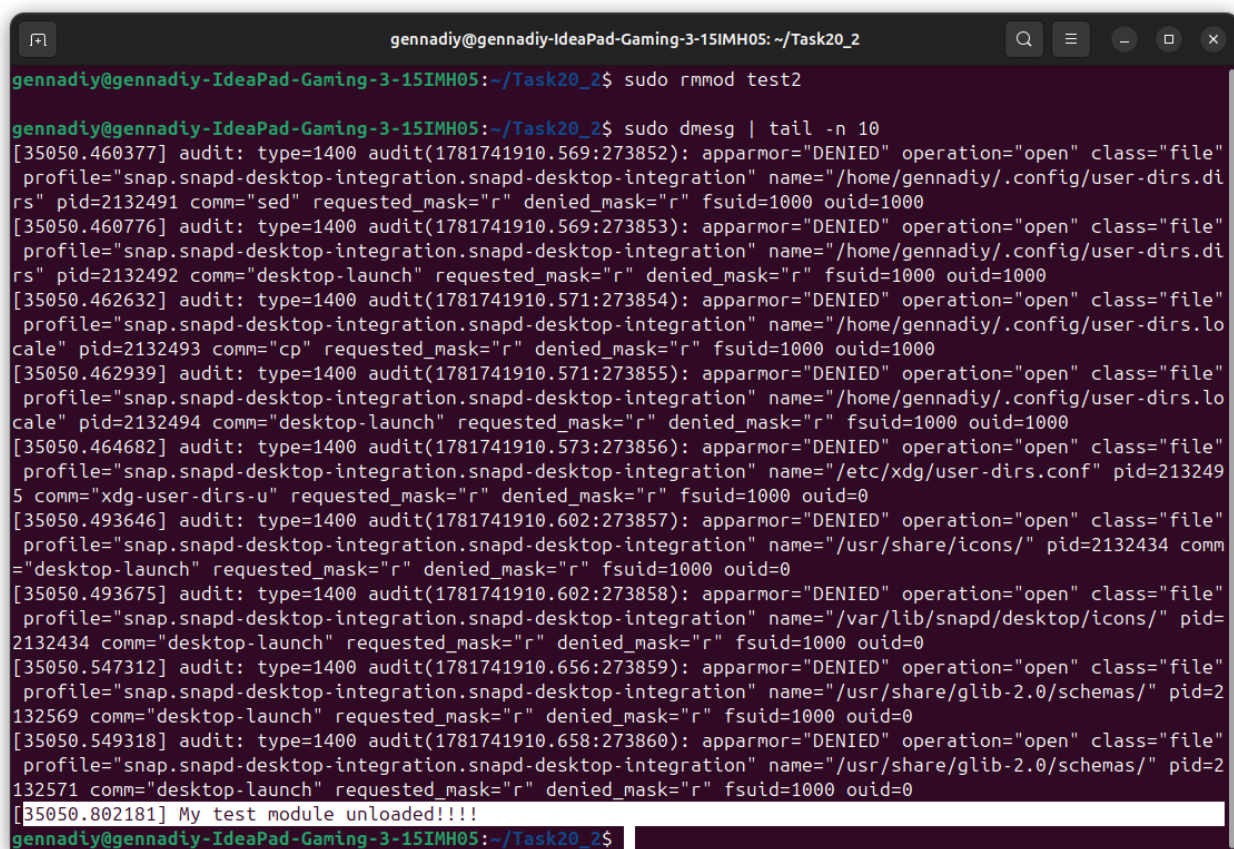
Выполняется тестирование на запись данных в procfs с помощью утилиты echo и чтение из sysfs командой cat. Для чтения из procfs использовалась альтернативная проверка командой head.

```

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ cat /sys/kernel/kobject_test/test_string
Hello!
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ echo "Hi" > /proc/test
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ cat /proc/test
cat: /proc/test: Неправильный адрес
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ cat /proc/test
cat: /proc/test: Неправильный адрес
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ head -n 1 /proc/test
head: /proc/test: невозможно переместить указатель позиции на -12: Недопустимая операция смещения
Hi
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ echo "Hihihih" > /proc/test
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ head -n 1 /proc/test
head: /proc/test: невозможно переместить указатель позиции на -7: Недопустимая операция смещения
Hihihih
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ cat /proc/test
cat: /proc/test: Неправильный адрес

```

Рисунок 3 — тестирование модуля ядра



```

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo rmmod test2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo dmesg | tail -n 10
[35050.460377] audit: type=1400 audit(1781741910.569:273852): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.dirs" pid=2132491 comm="sed" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[35050.460776] audit: type=1400 audit(1781741910.569:273853): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.dirs" pid=2132492 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[35050.462632] audit: type=1400 audit(1781741910.571:273854): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.locale" pid=2132493 comm="cp" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[35050.462939] audit: type=1400 audit(1781741910.571:273855): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.locale" pid=2132494 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[35050.464682] audit: type=1400 audit(1781741910.573:273856): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/etc/xdg/user-dirs.conf" pid=2132495 comm="xdg-user-dirs-u" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[35050.493646] audit: type=1400 audit(1781741910.602:273857): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/icons/" pid=2132434 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[35050.493675] audit: type=1400 audit(1781741910.602:273858): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/var/lib/snapd/desktop/icons/" pid=2132434 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[35050.547312] audit: type=1400 audit(1781741910.656:273859): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/glib-2.0/schemas/" pid=2132569 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[35050.549318] audit: type=1400 audit(1781741910.658:273860): apparmor="DENIED" operation="open" class="file"
  profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/glib-2.0/schemas/" pid=2132571 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[35050.802181] My test module unloaded!!!!
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$

```

Рисунок 4 — лог при выгрузке

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/Task20_2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo insmod test2.ko
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo dmesg | grep "Major"
[36557.990339] Major = 504
[36673.236712] Major = 504
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo mknod /dev/test c 504 0
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ sudo chmod 666 /dev/test
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ head -n 1 /dev/test
Hello!
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ cat /dev/test
cat: /dev/test: Неправильный адрес
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ head -n 1 /dev/test
Hello!
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ echo "AAAAAA" > /dev/test
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$ head -n 1 /dev/test
AAAAAA
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20_2$
```

Рисунок 5 — тестирование символьного устройства